ANALYZING THE CORRELATION BETWEEN SURPRISING MATCH RESULTS AND X TWEET ENGAGEMENT LEVELS FOR ENGLISH PREMIER LEAGUE CLUBS IN THE 2025-2026 SEASON

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INTRODUCTION

- **Research question:** Do surprising Premier League match outcomes drive engagement spikes on clubs' official X (twitter) accounts?
- **Approach:**
  - Pre-match betting odds → expected result
  - Actual result vs. expected → classify as expected, surprise_win, surprise_draw, or surprise_loss (per club)
  - Engagement lift = ratio of post-match to pre-match average tweet engagement (likes + retweets + replies + quotes)
- **Scope:** All 20 clubs, 2025/26 season, ~330 finished matches analyzed

DATA SOURCE

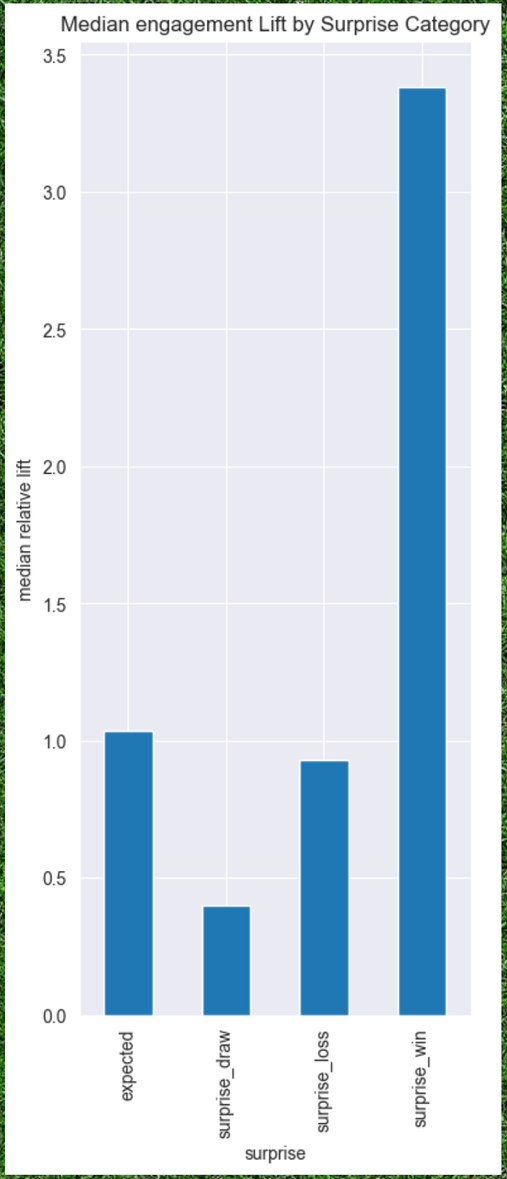
Data source #	1	2	
Name	Premier League Match Statistics (2025–2026 Season)	Premier League Betting Odds (Pre-Match Market Data)	Of En
Source URL	https://www.football-data.org/	Downloaded from https://www.football-data.co.uk/englandm.php	Of ht ht – t
Type	API	File	We of
		Pre-match betting odds	Tw

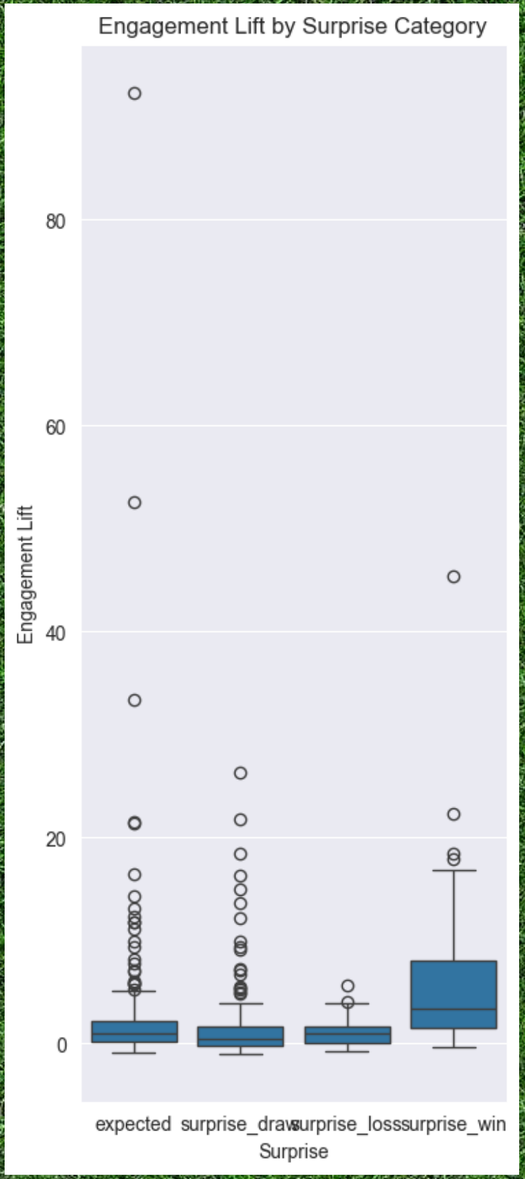
- Unification challenge: Each source uses different team names ("Nott'm Forest" vs "Nottingham" vs "NFFC") → built a mapping dictionary
- Storage: SQLite for joins, JSON for raw cache, CSV for odds
- Final dataset: 591 club-match rows after filtering clubs that didn't post around kickoff

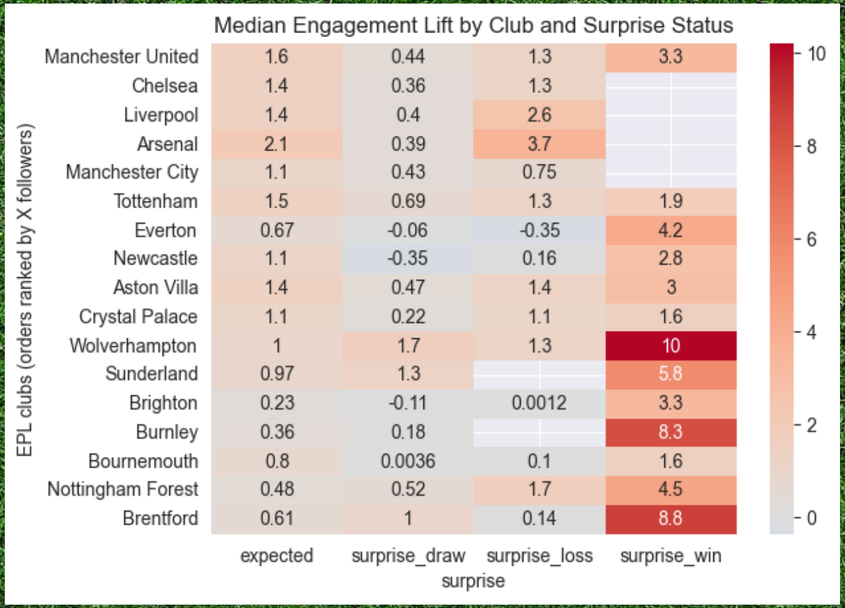
RESULT 1

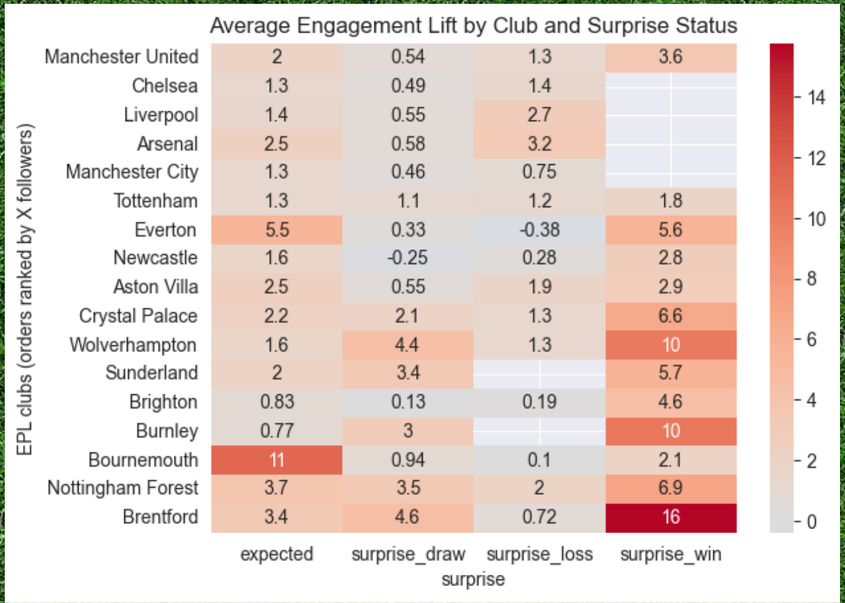
SURPRISING WINS DRIVES SIGNIFICANT ENGAGEMENT LIFT BUT NOT SURPRISING LOSSES

- Median lift by category:
  - Expected result: 1.0×
  - Surprise draw: 0.4×
  - Surprise loss: 0.9×
  - Surprise win: 3.4×
- Surprise wins produce the only meaningfully elevated engagement response
- Surprise losses and draws show no positive lift vs. expected results





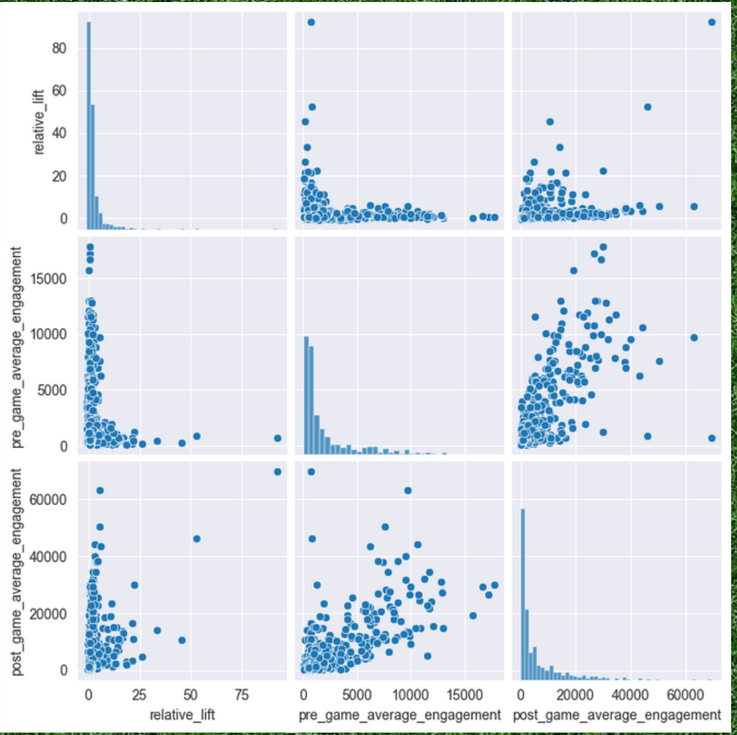


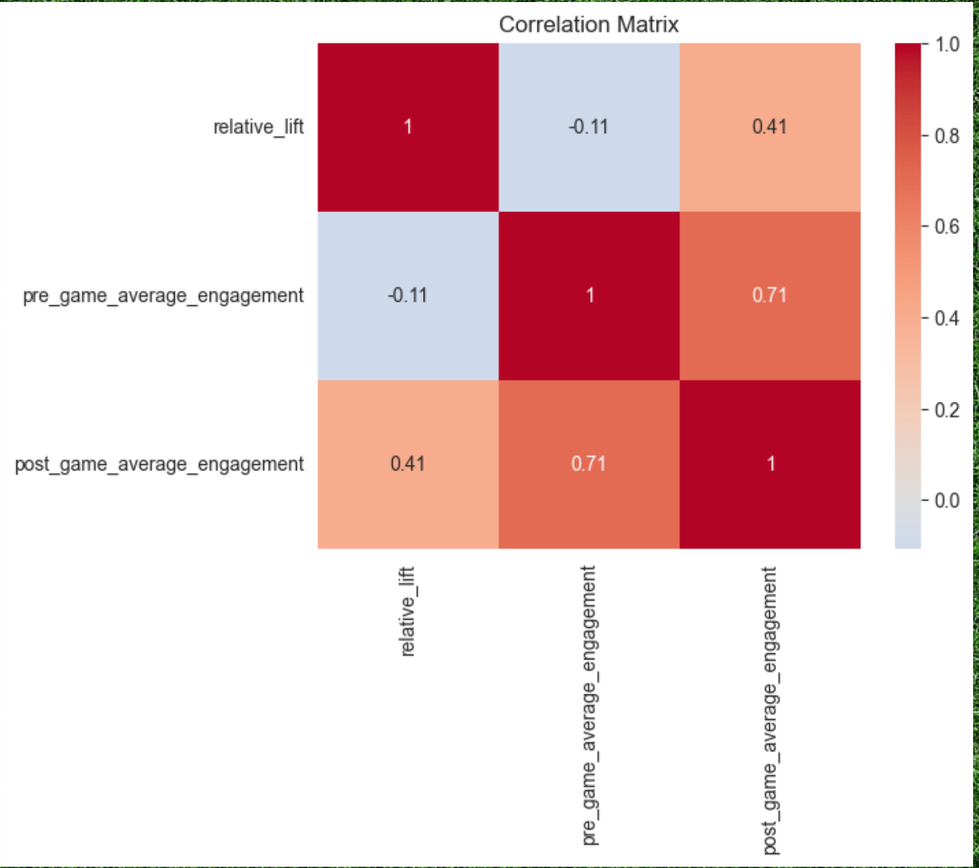


RESULT 2

SURPRISE WINS GENERATE THE LARGEST ENGAGEMENT SPIKES ACROSS SMALL CLUBS.

- Largest surprise_win lifts: Wolverhampton (10×), Brentford (8.8×), Burnley (8.3×)
- Smallest surprise_win lifts: Manchester United (3.3×), Arsenal (3.7×), Liverpool (2.6×)
- Two forces at work:Top clubs are rarely underdogs → fewer observations, and when they win it's less "surprising"
- Smaller clubs have lower baseline engagement → more room for relative lift





RESULT 3

POST-MATCH ENGAGEMENT DRIVES MOST OF THE VARIATION IN RELATIVE LIFT.
PRE- AND POST-MATCH ENGAGEMENT LEVELS ARE STRONGLY CORRELATED.

- relative_lift vs pre = -0.11
- relative_lift vs post = 0.41
- pre vs post = 0.71

SUMMARY OF RESULTS

Unexpected EPL match outcomes, espacially surprise wins, are associated with substantially higher engagement lift on official club X accounts. Relative lift analysis shows this effect persists after controlling for baseline engagement. While surprise losses also increase engagement moderately, surprise draws have weaker effects. Engagement responses vary by club size but follow consistent patterns overall.

CHALLENGES

- **X API cost:** Single full-season scrape burned ~$55. Built a per-match JSON cache so re-runs are free
- **Three data sources, three naming conventions:** "Nott'm Forest" / "Nottingham" / "NFFC" all mean the same club. Required a unification layer before any analysis could happen
- **Measurement validity:** Clubs that didn't post around kickoff were silently contaminating the "expected" category with zero-division artifacts. Required filtering on both pre- and post-window averages being nonzero
- **Scale comparability:** Raw differences scale with club size (Man Utd's average tweet >> Burnley's), so switched from absolute difference to relative lift ratio for cross-club comparability

THANK YOU